FUNCTIONAL DEPENDENCY

There exists a relation such that an attribute determines another attribute is called as functional dependency.

EX: PRODUCT

PID	PNAME	PRICE	DISCOUNT
A001	Iphone16	69999	1999
A002	SamsungS24	129999	2999
A003	Oneplus11	59999	999
A004	Iphone14	78999	1999

NOTE: In the above example if we need details of any product, we directly depend on **PID** this is the example of functional dependency.

TYPES OF FUNCTIONAL DEPENDENCY

1) Total Functional Dependency
2) Partial Functional Dependency
3) Transitive Functional Dependency

1) TOTAL FUNCTIONAL DEPENDENCY

There exists a relation such that all the attributes in a relation are determined by **key attribute** is called as total functional dependency.

EX: **EMP**

EMPNO	ENAME	SAL	JOB	HIREDATE	MGR
7121	Smith	1200	Clerk	01-JAN-1982	7124
7122	Miller	1500	Manager	15-DEC-1982	7124
7123	Scott	5500	Salesman	22-MAY-1988	7122
7124	James	3500	Clerk	23-JUL-1985	7123

- In the above example table, we have only one key attribute i.e., **EMPNO**. So, all the attributes are determined by **EMPNO**.
- If this relation exists in the table, it is said to have **Total Functional Dependency**.

2) PARTIAL FUNCTIONAL DEPENDENCY

There exists a relation such that an attribute determines another attribute which is a part of composite key attribute.

EX: **EMP**

EMPNO	ENAME	SAL	PHNO	HIREDATE	MGR
7121	Smith	1200	9876543210	01-JAN-1982	7124
7122	Miller	1500	9876543219	15-DEC-1982	7124
7123	Scott	5500	9876543218	22-MAY-1988	7122
7124	James	3500	9876543217	23-JUL-1985	7123

- In the above example table, we have 2 key attributes i.e., **EMPNO** and **PHNO.**
- Now the dependency is partially determined by only one key attribute.
- If the table has this relation it is said to have **Partial Functional Dependency.**

3) TRANSITIVE FUNCTIONAL DEPENDENCY

There exists a relation such that an attribute determines another attribute which in turn determined by key attribute.

EX: **EMP**

EMPNO	ENAME	SAL	DEPTNO	DNAME	LOC
7121	Smith	1200	20	Sales	Chicago
7122	Miller	1500	20	Sales	Chicago
7123	Scott	5500	10	Accounting	Dallas
7124	James	3500	30	Operations	New York

- In the above example table, **DEPTNO** belongs to dept details.
- If we need to retrieve **LOC,** we depend on **DEPTNO** and if we need **DEPTNO** we depend on **EMPNO.**
- If this relation exists in the table, then it is said to have **Transitive Functional Dependency.**